

**A
PROJECT REPORT
ON
“ArtNexus: AI-Enabled Online Art Exhibition Platform “
At**

‘SDK Infitech Pvt. Ltd ‘

By

Ms. Vaishnavi Jadhav

Under the guidance of

Prof. Mukund Thakur

Submitted to

‘Savitribai Phule Pune University ‘

**In partial fulfilment of the required for the award of the degree Master of
Computer Application (M.C.A)**

A.Y. 2025-2026

Through



**Purandhar Technical Education Society's
Pune Cambridge Institute of Management & Computer Application, Pune – 46**

June 01, 2026

CERTIFICATE

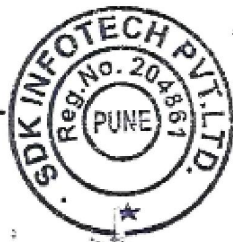
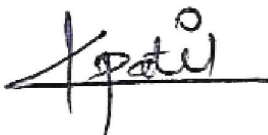
To Whom It May Concern:

Darshan Kalyan Jagdale has successfully completed her Internship Training as an **Intern Java Developer** at SDK INFOTECH PVT. LTD. from **December 01, 2025** to **June 01, 2026**.

During her tenure, she was involved in client related assignments. She has completed her assigned work.

We thank you for your valuable contributions during your association with us and we wish you all the best in your future endeavors.

Regards,



Head – Human Resources
SDK Infotech Pvt. Ltd.



PURANDHAR TECHNICAL EDUCATION SOCIETY'S
**PUNE CAMBRIDGE INSTITUTE OF
MANAGEMENT AND COMPUTER APPLICATION**

Affiliated to Savitribai Phule Pune University
Recognized by A.I.C.T.E., New Delhi & Govt of Maharashtra
DTE CODE: 6992



90285917 25 / 26 / 28



www.pcimca.in



info@pcmca.in



Sr. No -28/6, behind Ambegaon Pathar
Police Chowki, Ambegaon Pathar, Ambegaon Bk
Pune, Maharashtra 411046

CERTIFICATE

This is to certify that Ms. Vaishnavi Ram Jadhav, Roll No. MCA24057, student of "Pune Cambridge Institute of Management & Computer Application, Pune - 46" has completed her Project on " Title "and has submitted the Project report in partial fulfillment of the Master of Computer Application Course of the Savitribai Phule Pune University (SPPU) for the Semester IV Academic Year 2025-2026. She has worked under our guidance, and the said report is based on bonafide information.

Prof. Sonali Gaikwad
Project Guide

Prof. Shubham Jambhale
HOD, MCA

Dr. Manohar Karade
Director

Internal Examiner

External Examiner

PCIMCA

DECLARATION

I, **Vaishnavi Ram Jadhav** the undersigned, hereby declare that the project titled “**ArtNexus: AI-Enabled Online Art Exhibition Platform**”, being submitted for the award of the degree of **Master of Computer Application** by me to **Pune Cambridge Institute of Management & Computer Application, Pune – 46**, affiliated to **Savitribai Phule Pune University, Pune**, is the result of an independent work carried out under the guidance of **Prof. Mukund Thakur** and is my original work. Further, I declare that this project has not been submitted to this or any institution for the award of any degree.

(Signature)

Vaishnavi Ram Jadhav

Place : _____

Date : _____

ACKNOWLEDGEMENT

I would like to take this opportunity to thank the people who directly or indirectly enabled me to work on this project. At the outset, I would like to thank the entire staff of Pune Cambridge Institute of Management & Computer Application, Pune -46, both the teaching and non-teaching staff, for their continuous and unrelenting efforts in helping us with our project.

Without them, these efforts would not have given us expected results. I started this project as part of the curriculum Master of Computer Application (Sem IV). During this work, I have gained both practical as well as theoretical knowledge of great significance. Now, after the completion of the project, I feel very pleased to present this project report. Also, I would like to take this opportunity to thank all those who helped us to complete the project. I would especially like to thank our Director Dr. Manohar Karade & our head of department Prof. Shubham Jambhale for their encouragement & continuous support. I am extremely grateful to our Internal Project Guide Prof. Mukund Thakur for his constant and valuable encouragement. It is their inexorable direction and profound experience that motivated and inspired us. Finally, my heartfelt appreciation to my teammates for their valuable advice and timely support. Expressing our hearty gratitude. Thank you.

ABSTRACT

ArtNexus is an AI-enabled online art exhibition platform developed as a web-based application to provide a modern digital space where artists can showcase their artwork and users can explore, interact with, and purchase artworks through an enquiry-based system. The platform addresses the limitations of traditional physical art exhibitions by removing geographical, time, and cost-related barriers.

The implemented website provides a home page, art gallery, category navigation, login and registration pages, artwork cards, price filters, sorting options, wishlist/cart buttons, contact information, and an ArtNexus chatbot widget. The live gallery currently shows featured artworks such as Color Burst, Nature's Escape, Silent Expression, and Future Vision.

The system is built using a three-tier architecture consisting of a React-based frontend, a Python Flask backend, and a database layer. It includes User, Artist, Admin, Artwork, Enquiry, Category, and AI Chatbot modules. Key functions include artwork browsing, category filtering, price-based filtering, enquiry management, email notifications, image hosting, and admin-side CRUD operations.

Keywords: ArtNexus, Online Art Exhibition, AI Chatbot, Django, HTML,CSS,JAVASCRIPT,BOOSTRAP, MySQL, Artwork Enquiry System, Admin Panel

INDEX

SR. NO.	CONTENT	PAGE NO.
Chapter 1	Introduction	
1.1	Problem Statement	
1.2	Objectives	
1.3	Scope	
Chapter 2	Design	
2.1	System Architecture	
2.2	Database Design	
Chapter 3	Implementation	
3.1	Frontend Development	
3.2	Backend Development	
3.3	Integration	
Chapter 4	Testing	
4.1	Test Cases	
4.2	Results	
Chapter 5	Conclusion	
5.1	Summary	
5.2	Future Enhancement	
Chapter 6	References	
Chapter 7	Appendices	
Chapter 8	Annexure – Progress Sheet	

CHAPTER 1: INTRODUCTION

In the modern digital era, technology plays a transformative role in converting traditional processes into efficient, scalable digital solutions. The art industry can benefit significantly from digital platforms, as artists can showcase their creative work to a wider audience without the restrictions of physical galleries, venue costs, fixed schedules, and geographical limitations.

ArtNexus is an AI-enabled online art exhibition platform designed to provide a comprehensive digital space where artists present artwork and art enthusiasts explore, discover, and interact with creative works. The platform supports diverse artwork categories including Digital Art, Abstract Art, Portrait Art, Landscape Art, Sculptures, Serigraphs, Prints, Paintings, Street Art, Visual Art, and Conceptual Art.

The live system provides a home page with the tagline 'Discover & Buy Unique Art', an art gallery, category navigation, login and registration pages, a contact page, featured artwork cards, and an ArtNexus chatbot widget for user assistance. The platform is deployed and accessible at <https://artnexus.onrender.com>.

ArtNexus bridges the gap between traditional art exhibition culture and modern digital accessibility. The platform enables art enthusiasts to discover, explore, and enquire about artworks from anywhere in the world, and simultaneously enables artists to gain visibility beyond geographic and temporal limitations.

1.1 Problem Statement

Traditional art exhibitions are constrained by physical location, time schedules, high setup costs, and limited audience reach. Artists from small cities may not get enough opportunities to showcase their work to audiences outside their region, while buyers are limited to exhibitions available nearby. This creates a significant imbalance in exposure and opportunity within the art ecosystem.

Existing online art platforms provide visibility but often lack intelligent interaction, structured enquiry handling, affordable exposure for independent artists, and a centralized admin system. Users may struggle to find artworks matching budget, style, or category preferences without a guided discovery mechanism.

Existing platforms, while providing some digital presence, often suffer from several shortcomings. Most platforms lack intelligent interaction features, AI-powered assistance, or structured communication channels between buyers and artists. Users often struggle to find artworks that match their preferences and budget, and there is no centralized system for managing enquiries efficiently.

Furthermore, there is a lack of properly designed admin systems that allow platform administrators to manage artworks, artists, and enquiries from a single dashboard. Artists also face difficulty in getting their work properly listed, categorized, and made visible to the right audience.

Therefore, there is a clear need for a system that provides an easy, interactive, and intelligent platform for art exhibition, user engagement, enquiry management, and comprehensive admin control — which is precisely what ArtNexus addresses.

Existing platforms like Saatchi Art and Artsy primarily target international artists, leaving Indian regional artists without affordable listing options and local visibility. No existing platform provides buyers with an AI-driven conversational interface where they can search artworks in natural language. Physical gallery logistics such as artwork insurance, packaging, and transportation are expensive barriers that a digital platform completely eliminates. Traditional platforms lack structured enquiry tracking, and small and independent artists struggle to compete with established names on generic platforms that do not offer category-specific or region-specific visibility tools.

1.2 Objectives

The ArtNexus project was developed with the following primary and secondary objectives:

- To develop a web-based platform for displaying and exploring diverse categories of artwork.
- To allow users to browse artworks by category, view details, and apply price-based filters.
- To enable users to submit purchase enquiries through a structured enquiry system.
- To provide artists with a platform to showcase artwork and connect with potential buyers.
- To implement AI chatbot assistance for navigation, queries, and artwork discovery.
- To provide a secure admin panel for managing artworks, artists, categories, users, and enquiries.
- To implement secure authentication and protect admin-only functionality.
- To deploy a scalable and maintainable web application using modern web technologies.
- To maintain a wishlist and cart system so users can save artworks for future reference without immediate purchase commitment.
- To provide an automated email notification system that informs users about status updates of their submitted enquiries.
- To ensure platform responsiveness across all device sizes including mobile, tablet, and desktop using the Bootstrap grid system.
- To store artwork images securely on Cloudinary CDN for faster load times and reduced server storage dependency.
- To allow the admin to track and manage enquiry status (Pending / Approved / Rejected) from a single centralized dashboard.
- To implement input validation on both the frontend and backend layers to ensure data integrity and prevent invalid submissions.

1.3 Scope of the Project

The scope of the ArtNexus platform covers the development and deployment of a comprehensive online artwork exhibition and management system that connects art enthusiasts with artists through a centralized digital platform. The system aims to simplify artwork discovery, artist promotion, and communication between users and administrators.

User Scope

- Users can create and manage their accounts through a secure registration and login system.
- Users can browse artworks displayed on the platform without technical knowledge.
- Users can search artworks using artwork title, category, artist name, and keywords.
- Users can filter artworks based on category, artwork type, and price range.
- Users can view detailed artwork information including image, title, description, price, artist information, and category.
- Users can submit enquiries regarding artworks through the enquiry form.
- Users can receive guidance and recommendations through the AI-powered chatbot.
- Users can access the platform from different devices using responsive web design.
- Users can explore featured and recently added artworks.
- Users can view artist profiles and learn about their artistic background.

Artist Scope

- The platform provides visibility to artists by showcasing their artworks online.
- Artist details including biography, specialization, achievements, and contact information can be maintained.
- Multiple artworks can be associated with a single artist profile.
- Artists can gain exposure to a larger audience through the online exhibition platform.
- The system supports categorization of artworks based on artist specialization and artwork type.
- Artist information can be updated and managed through administrative controls.

Administrative Scope

- Administrators can securely log in and access management functionalities.
- Administrators can perform Create, Read, Update, and Delete (CRUD) operations on artworks.
- Administrators can manage artist records and maintain updated artist information.
- Administrators can create and manage artwork categories.
- Administrators can review and respond to user enquiries.
- Administrators can monitor platform activities and maintain content quality.
- Administrators can manage artwork images through Cloudinary integration.
- Administrators can receive notifications regarding enquiries and important system events.
- Administrators can control platform content and prevent unauthorized modifications.

Artificial Intelligence Scope

- The AI chatbot assists users in navigating the platform effectively.
- The chatbot provides instant responses to frequently asked questions.
- Users can receive artwork recommendations based on budget preferences.
- The chatbot helps users locate specific categories and artworks.

- AI integration improves overall user engagement and user experience.
- Future enhancements can include personalized recommendations based on user behavior and browsing history.

Database Scope

- The system utilizes a MySQL relational database for structured data storage.
- Database tables are normalized up to Third Normal Form (3NF) to reduce redundancy.
- Separate tables are maintained for users, artists, artworks, categories, enquiries, and chatbot interactions.
- Foreign key relationships ensure data consistency and integrity.
- Database indexing improves query performance and retrieval speed.
- Backup and recovery strategies can be implemented to prevent data loss.

Security Scope

- User authentication mechanisms protect access to sensitive functionalities.
- Passwords are stored using secure encryption and hashing techniques.
- Role-based access control separates administrative and user privileges.
- Unauthorized access attempts are restricted through authentication middleware.
- Input validation techniques help prevent malicious data entry.
- Secure session management enhances overall platform security.
- Protection measures can be implemented against common web vulnerabilities such as SQL Injection and Cross-Site Scripting (XSS).

Technology Scope

- Frontend development is implemented using React.js.
- Backend services are developed using Python Flask framework.
- RESTful APIs facilitate communication between frontend and backend components.
- Cloudinary is used for efficient image storage and delivery.
- MySQL serves as the primary relational database management system.
- Git and GitHub support version control and collaborative development.
- Render cloud platform provides hosting and deployment services.

Performance and Usability Scope

- The application is designed to provide fast and responsive user interactions.
- Optimized image loading improves website performance.
- Responsive layouts ensure compatibility across desktop, tablet, and mobile browsers.
- User-friendly interfaces enhance navigation and usability.
- Efficient database queries reduce response time and improve scalability.

Deployment Scope

- The platform is deployed on the Render cloud platform.
- Users can access the system through a publicly available URL.
- No local software installation is required for accessing the application.
- Cloud deployment ensures high availability and easy maintenance.
- Future migration to dedicated cloud infrastructure can be supported.

Future Enhancement Scope

- Integration of online payment gateways for artwork purchases.
- Development of Android and iOS mobile applications.
- Real-time chat between buyers and artists.
- Artist self-registration and profile management portal.
- Multi-currency and international pricing support.
- Bulk artwork upload functionality.
- Social media integration for artwork sharing.
- NFT and blockchain-based artwork ownership verification.
- Advanced recommendation systems using machine learning algorithms.
- Analytics dashboard for artists and administrators.

Project Limitations

- Online payment functionality is currently not implemented.
- Native mobile applications are not available in the current version.
- Real-time buyer-artist communication is not supported.
- NFT marketplace features are excluded from the current implementation.
- Multi-language support is not available.
- Artists cannot independently register and upload artworks.
- Bulk import/export of artworks is not supported.
- Advanced personalization features are reserved for future versions.

1.4 Methodology (SDLC Approach)

The development of the ArtNexus platform followed the Software Development Life Cycle (SDLC) approach. SDLC provides a structured framework for planning, designing, developing, testing, deploying, and maintaining software systems. The methodology ensures that the project is completed systematically while meeting user requirements and quality standards.

1. Planning Phase

The planning phase was the initial stage of the project where the objectives, scope, and feasibility of the system were identified. The primary goal was to develop an online platform that enables users to explore artworks, discover artists, and interact with an intelligent chatbot. The activities performed during this phase include:

- Identification of project objectives and goals.
- Understanding the needs of art enthusiasts and artists.
- Defining project scope and limitations.
- Conducting feasibility analysis.
- Identifying hardware and software requirements.
- Preparing project timelines and milestones.
- Assessing risks and resource availability.

The outcome of this phase was a clear roadmap for project development.

2. Requirement Analysis and SRS Preparation

In this phase, detailed requirements were collected and analyzed. User expectations and business requirements were documented in the Software Requirement Specification (SRS) document. The requirements gathered included:

Functional Requirements:

- User Registration and Login
- Artwork Browsing
- Category-Based Filtering
- Price-Based Filtering
- Artwork Detail View
- Artist Information Display
- Enquiry Submission
- AI Chatbot Assistance
- Admin Dashboard
- Artwork Management
- Artist Management
- Category Management

Non-Functional Requirements:

- Security
- Scalability
- Reliability
- Performance
- Usability
- Maintainability

The SRS document served as the foundation for design and development activities.

3. System Design Phase

The design phase focused on converting requirements into technical specifications and architectural components.

Architecture Design:

The application follows a three-layer architecture:

- Presentation Layer (Frontend)
- Business Logic Layer (Backend)
- Data Layer (Database)

This architecture improves maintainability and scalability. The database schema was designed using normalization principles to eliminate redundancy and maintain consistency. Major entities include Users, Artists, Artworks, Categories, Enquiries, and Chatbot Logs. Relationships between tables were established using primary and foreign keys.

UI wireframes and screen layouts were prepared to ensure easy navigation, responsive design, user-friendly interaction, and consistent visual appearance. The design phase also included planning page structures, menus, forms, and chatbot interfaces.

4. Implementation Phase

The implementation phase involved converting the design into a fully functional web application. The user interface was developed using React.js, Bootstrap 5, HTML5, CSS3, and JavaScript. The backend was built using Python Flask, and all components were integrated to form a fully functional full-stack application.

5. Testing Phase

The testing phase involved verifying that all components work correctly, individually and together. Unit testing, integration testing, system testing, and user acceptance testing were performed. A total of 20 test cases were executed, and all returned PASS results, confirming the quality and reliability of the system.

6. Deployment Phase

The completed application was deployed on the Render cloud platform. The React frontend and Flask backend are hosted as separate services on Render. The MySQL database is managed through Render's managed database service, and Cloudinary CDN is used for media storage. The live platform is accessible at <https://artnexus.onrender.com>.

CHAPTER 2: DESIGN

2.1 System Architecture

The ArtNexus platform is designed using a Three-Tier Architecture, which provides a clear separation between the user interface, business logic, and data storage layers. This architectural approach improves system scalability, maintainability, security, and flexibility. Each layer performs a specific set of responsibilities and communicates with other layers through well-defined interfaces.

The three-tier architecture enables independent development and maintenance of system components. Changes made in one layer have minimal impact on other layers, making the system easier to upgrade and extend in the future.

Tier 1: Presentation Layer (Frontend)

The Presentation Layer serves as the interface between the users and the system. It is responsible for displaying information, collecting user input, and communicating with the backend services. The frontend of ArtNexus is developed using React.js, Bootstrap 5, HTML5, CSS3, and JavaScript. This layer provides an interactive and user-friendly interface that allows users to navigate the platform efficiently.

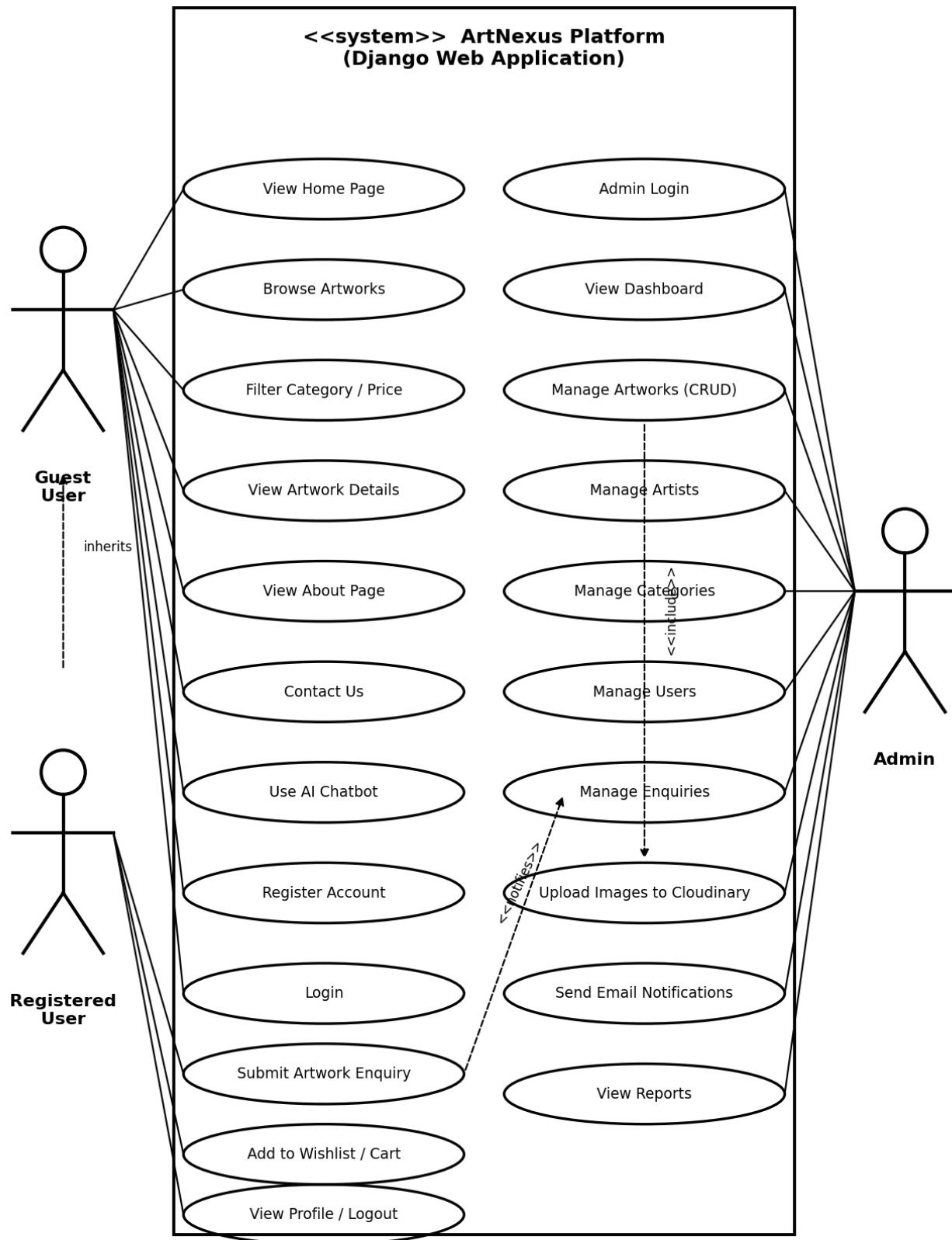
Responsibilities of the Presentation Layer:

- Displaying artwork listings and categories.
- Showing artwork details including images, descriptions, prices, and artist information.
- Providing user registration and login interfaces.
- Allowing users to submit enquiries.
- Offering chatbot interaction screens.
- Displaying artist profiles and information.
- Providing responsive layouts for different screen sizes.
- Managing user navigation throughout the application.

Features of the Presentation Layer:

- Component-based UI architecture using React.js.
- Responsive design through Bootstrap.
- Fast rendering and dynamic updates.
- Reusable user interface components.
- Improved user experience and accessibility.
- Cross-browser compatibility.

Use Case Diagram - ArtNexus Platform



Tier 2: Application Layer (Backend)

The Application Layer acts as the bridge between the frontend and the database. It contains all business logic and processing rules required for the functioning of the platform. The backend is developed using Python, Flask Framework, and a RESTful API architecture. The Application Layer receives requests from the frontend, processes the requests according to business rules, interacts with the database, and returns appropriate responses.

Responsibilities of the Application Layer:

- User authentication and authorization.
- Processing user registration and login requests.
- Managing artworks and categories.
- Handling artist information.
- Processing enquiry submissions.
- Managing chatbot interactions.
- Communicating with Cloudinary for image storage.
- Validating user input and request data.
- Implementing security and access control mechanisms.

The backend exposes several RESTful API endpoints for User Registration, User Login, Artwork Listing, Artwork Detail, Category Management, Artist Management, Enquiry Submission, and Chatbot API.

Tier 3: Data Layer (Database)

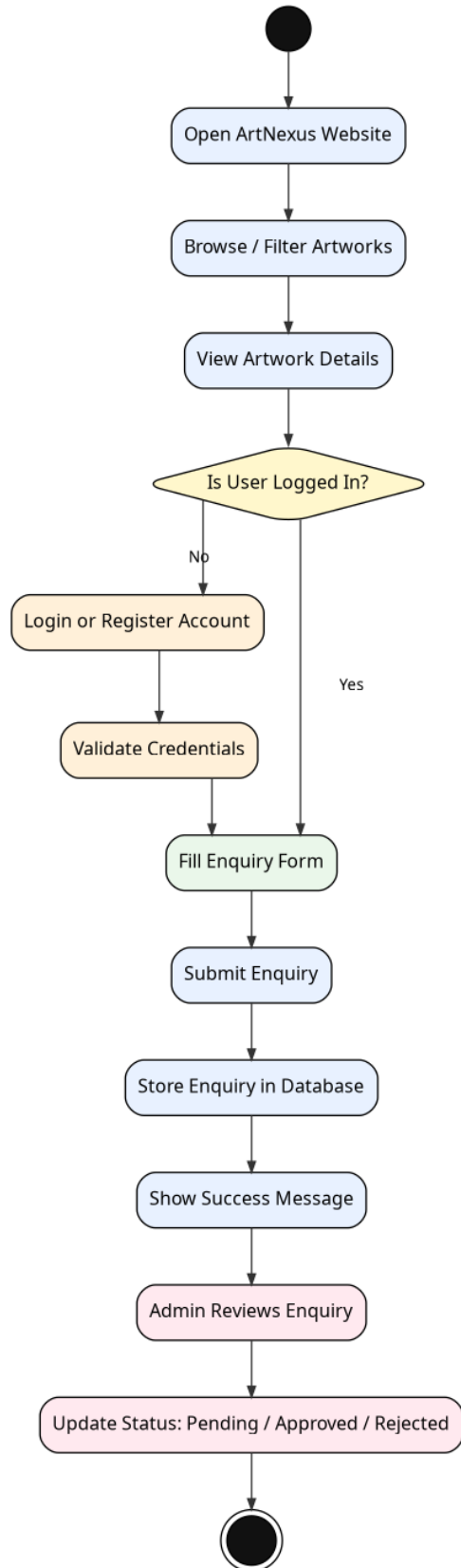
The Data Layer is responsible for storing, managing, and retrieving all application data. It serves as the central repository for the entire platform. ArtNexus uses MySQL Relational Database Management System. The database schema is designed using normalization principles to reduce redundancy and improve consistency.

The database contains several interconnected tables including Users, Artists, Artworks, Categories, Enquiries, Admin Records, and Chatbot Logs. The relational structure ensures consistency between entities and enables efficient retrieval of information across the platform.

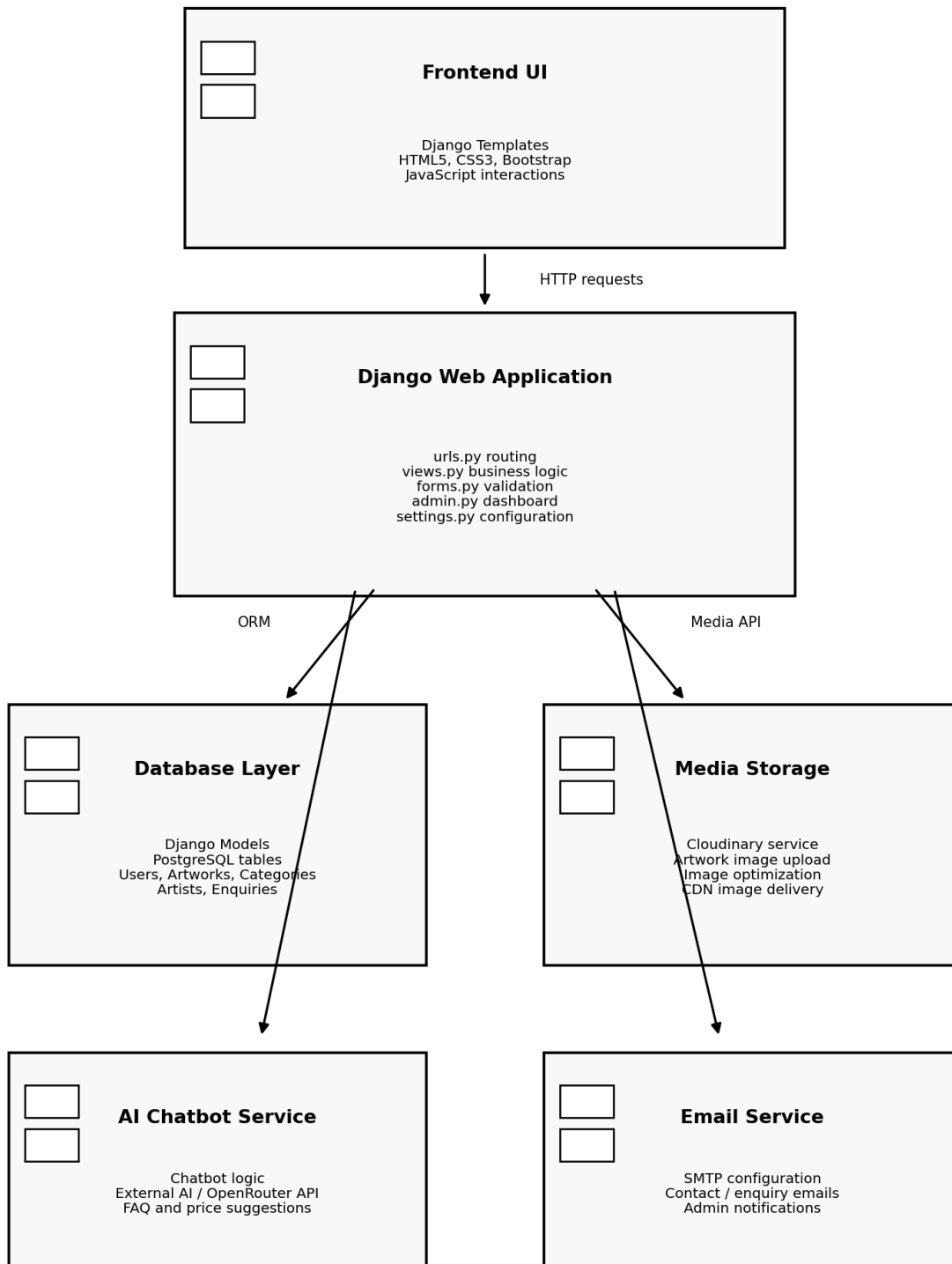
Database Features:

- Primary Key and Foreign Key relationships.
- Normalization up to Third Normal Form (3NF).
- Data integrity constraints.
- Efficient indexing for faster queries.
- Secure storage of sensitive information.

Activity Diagram - Artwork Enquiry Flow



Component Diagram - ArtNexus Platform



Cloud Storage Integration

In addition to the database layer, ArtNexus integrates with Cloudinary for storing and managing artwork images. Cloudinary provides secure cloud-based image storage, automatic image optimization, fast image delivery through CDN, reduced server storage requirements, and efficient handling of high-resolution artwork images. This integration improves application performance and scalability while maintaining image quality.

System Architecture — API Endpoints Overview

The Django backend exposes RESTful API endpoints:

Endpoint	Method	Description
/api/register	POST	User registration
/api/login	POST	User authentication
/api/artworks	GET	Fetch all artworks (with filter params)
/api/artworks/<id>	GET	Fetch single artwork details
/api/arttype/<id>	GET	Fetch artworks by category
/api/enquiry	POST	Submit purchase enquiry
/api/admin/artworks	POST/PUT/DELETE	Admin artwork management
/api/admin/artists	POST/PUT/DELETE	Admin artist management
/api/admin/enquiries	GET/PUT	Admin enquiry management
/api/chatbot	POST	AI chatbot interaction
/api/wishlist	POST	Add artwork to user wishlist
/api/cart	POST	Add artwork to shopping cart

Module Design

The ArtNexus system is divided into the following functional modules:

- User Module: handles registration, login, gallery browsing, artwork details, filters, chatbot use, and enquiry submission.
- Artist Module: stores artist profile information such as name, contact, email, and biography, and links artists to artworks.
- Artwork Module: manages artwork title, price, category, medium/type, dimensions/orientation, description, images, and artist mapping.
- Admin Module: provides protected dashboard access for CRUD operations on artworks, artists, categories, and enquiry management.
- AI Chatbot Module: processes user questions, identifies intent using predefined patterns/keyword matching, and assists with navigation or price/category-based recommendations.

- Enquiry Module: captures user interest in artwork, stores message/status, and supports admin follow-up through email notification.

Frontend–Backend Communication:

The React frontend uses the Axios library to make asynchronous HTTP API calls (GET, POST, PUT, DELETE) to Flask backend endpoints. JSON is the standard data exchange format between all layers.

State Management:

React Context API is used to manage global states such as user authentication token and login status across all components, avoiding the need for prop drilling through multiple component levels.

CORS Configuration:

Flask-CORS middleware is configured on the backend to allow only the authorized frontend origin to make cross-origin requests, preventing unauthorized API access from external domains.

Environment Variables:

Sensitive configuration data such as database connection strings, Cloudinary API keys, and mail server credentials are stored as environment variables and are never hardcoded in source files.

2.2 Database Design

The database consists of the following major entities: Users, Artists, Categories, Artworks, and Enquiries. The relationship structure is: one artist can have many artworks; one category can contain many artworks; one user can submit many enquiries; and one artwork can receive many enquiries.

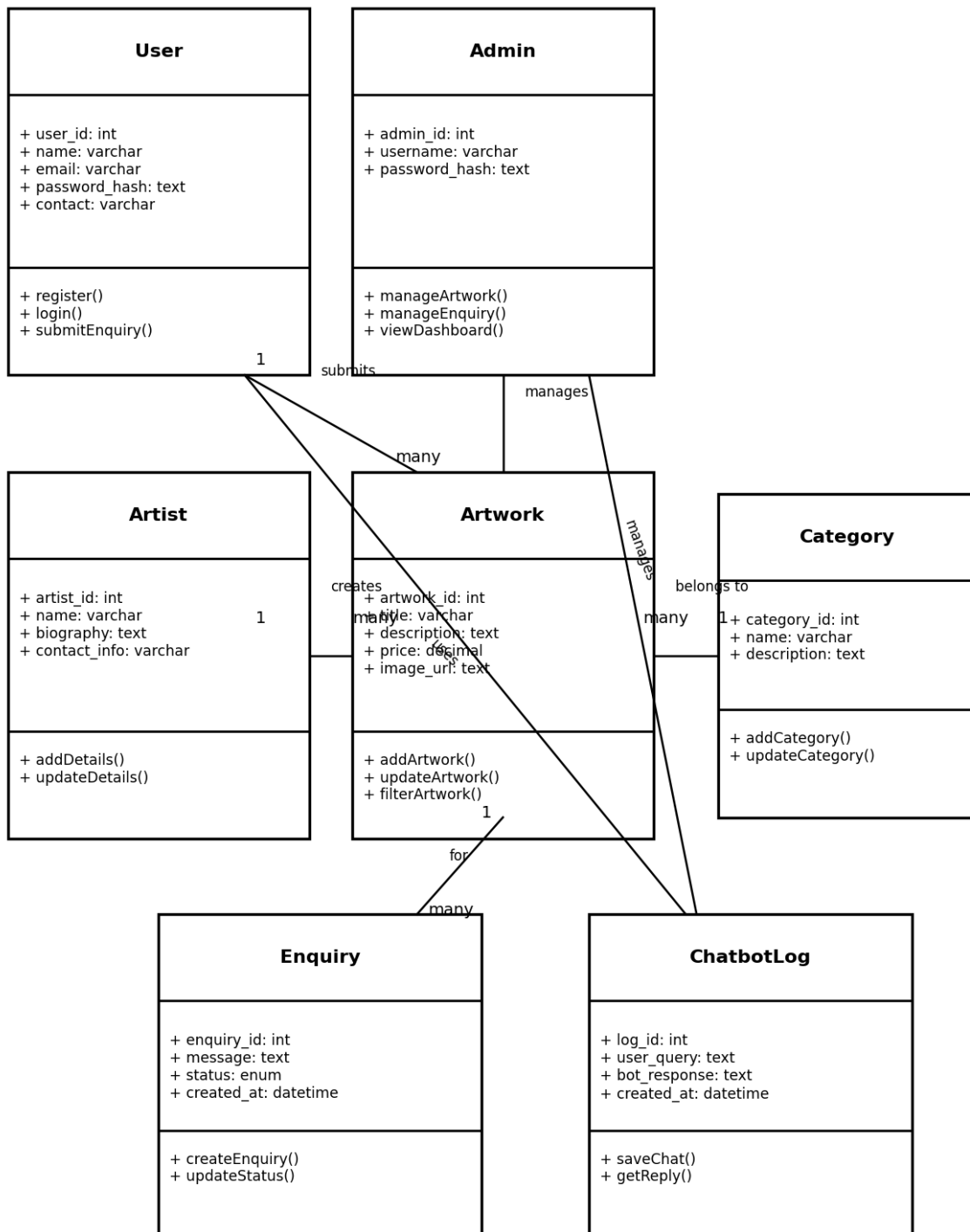
Normalization: The database schema follows Third Normal Form (3NF) to eliminate data redundancy. Category names are stored in a separate categories table and linked to artworks via `category_id` foreign key instead of repeating the category name in each artwork record.

Indexing: Primary keys are auto-incremented integers. Foreign key indexes on `artwork_id`, `user_id`, and `category_id` improve JOIN query performance during filtering and listing operations.

Password Security: Passwords in the users table are never stored in plain text. bcrypt hashing with an auto-generated salt is applied before every insertion into the database.

Status Field in Enquiries: The status field in the enquiries table uses predefined values — Pending, Approved, and Rejected — allowing the admin to filter and process enquiries based on their current state.

Class Diagram - ArtNexus Platform



Users Table

Field Name	Data Type	Constraints	Description
user_id	INT	PK, AUTO_INCREMENT	Unique User ID
full_name	VARCHAR(100)	NOT NULL	User Full Name
email	VARCHAR(150)	NOT NULL, UNIQUE	User Email Address
password	VARCHAR(255)	NOT NULL	bcrypt Hashed Password
contact_number	VARCHAR(15)	NULL	User Contact Number
role	VARCHAR(20)	DEFAULT 'user'	Role: user or admin
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Registration Date

Artists Table

Field Name	Data Type	Constraints	Description
artist_id	INT	PK, AUTO_INCREMENT	Unique Artist ID
artist_name	VARCHAR(100)	NOT NULL	Artist Full Name
biography	TEXT	NULL	Artist Biography
contact_info	VARCHAR(100)	NULL	Contact Details
email	VARCHAR(150)	NULL	Artist Email Address
profile_image	VARCHAR(255)	NULL	Cloudinary Image URL

Artworks Table

Field Name	Data Type	Constraints	Description
artwork_id	INT	PK, AUTO_INCREMENT	Unique Artwork ID
title	VARCHAR(200)	NOT NULL	Artwork Title
description	TEXT	NULL	Full Description
price	DECIMAL(10,2)	NOT NULL, > 0	Price in INR
image_url	VARCHAR(255)	NOT NULL	Cloudinary CDN URL
artist_id	INT	FK → artists	Linked Artist ID

Field Name	Data Type	Constraints	Description
category_id	INT	FK → categories	Linked Category ID
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Upload Date

Categories Table

Field Name	Data Type	Constraints	Description
category_id	INT	PK, AUTO_INCREMENT	Unique Category ID
category_name	VARCHAR(50)	NOT NULL	Category Name
description	TEXT	NULL	Category Description

Enquiries Table

Field Name	Data Type	Constraints	Description
enquiry_id	INT	PK, AUTO_INCREMENT	Unique Enquiry ID
user_id	INT	FK → users, NOT NULL	Linked User ID
artwork_id	INT	FK → artworks, NOT NULL	Linked Artwork ID
message	TEXT	NULL	Enquiry Message
status	ENUM	Pending/Approved/Rejected	Enquiry Status
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Submission Date

Entity Relationship Summary

Relationship	Type	Description
artists → artworks	One-to-Many	One artist can upload or be linked with multiple artworks.
categories → artworks	One-to-Many	One category can contain multiple artworks.
users → enquiries	One-to-Many	One user can submit multiple purchase enquiries.
artworks → enquiries	One-to-Many	One artwork can receive multiple enquiries from users.

2.3 API Design

Endpoint Group	Method	Endpoint	Description
Auth	POST	/api/register	Register a new user account
Auth	POST	/api/login	Authenticate user and return JWT token
Artworks	GET	/api/artworks	Retrieve all artworks with optional filters
Artworks	GET	/api/artworks/<id>	Retrieve a single artwork by ID
Artworks	POST	/api/artworks	Add a new artwork (admin only)
Artworks	PUT	/api/artworks/<id>	Update artwork details (admin only)
Artworks	DELETE	/api/artworks/<id>	Delete an artwork (admin only)
Categories	GET	/api/arttype/<id>	Retrieve artworks by category ID
Enquiries	POST	/api/enquiries	Submit a purchase enquiry
Enquiries	GET	/api/enquiries	Retrieve all enquiries (admin only)
Chatbot	POST	/api/chatbot	Send message and receive AI response
Wishlist	POST	/api/wishlist	Add artwork to user wishlist
Cart	POST	/api/cart	Add artwork to shopping cart

CHAPTER 3: IMPLEMENTATION

Implementation is the phase where the carefully crafted design is translated into a fully functional application. During this phase, all design blueprints, database schemas, API contracts, and UI wireframes are converted into working code. The ArtNexus platform is implemented as a full-stack web application combining a React.js frontend, a Flask REST API backend, a MySQL relational database, and Cloudinary cloud-based media storage. The application is deployed on Render's cloud hosting platform and is accessible at <https://artnexus.onrender.com>.

3.1 Frontend Development

3.1.1 Frontend Setup and Technology Stack

The React application is bootstrapped using Create React App (CRA), which provides a pre-configured webpack build pipeline, Babel transpiler for modern JavaScript, and an integrated development server. The key libraries and tools used in the frontend are listed in the table below:

Library / Tool	Version	Purpose
React.js	18.x	Core UI library for component-based rendering
React Router v6	6.x	Client-side routing and navigation management
Axios	1.x	HTTP client for REST API communication with Flask backend
Bootstrap 5	5.x	Responsive CSS grid and pre-built UI components
React Context API	Built-in	Global state management for authentication and cart
Cloudinary SDK	Latest	Frontend integration for media image URL generation

3.1.2 Key Frontend Modules and Pages

The ArtNexus frontend is organized into the following key pages and modules, each implemented as one or more React components:

Home Page (/)

The Home page serves as the landing page of the ArtNexus platform. It is designed to immediately communicate the platform's value proposition to visitors. The page is structured into the following sections:

- **Hero Banner:** A full-width banner with the tagline 'Discover & Buy Unique Art', a background image representing fine art, and two primary call-to-action buttons — 'Explore' (links to the Art Gallery) and 'Join Now' (links to Registration).
- **Why ArtNexus?:** A three-column section highlighting the platform's core advantages: Creative Platform, Global Reach, and Live Exhibitions.
- **Featured Artworks Carousel:** A horizontally scrollable carousel showcasing a curated selection of artworks. Each carousel item displays the artwork thumbnail, title, artist name, and price.

- **Navigation Bar:** Includes links to Home, About, Art, Contact, a Categories dropdown, Login, and Register.
- **Chatbot Widget:** A floating button labelled 'Chat with ArtNexus Bot' appears on all pages, including the Home page.
- **Footer:** Contains social media links, quick navigation links, and copyright information.

Art Gallery Page (/Artproductlist)

The Art Gallery page is the primary product listing page. It displays all available artworks in a responsive grid layout and provides several filtering and sorting controls:

- **Artwork Grid:** Each artwork is rendered as a card component showing the Cloudinary-hosted thumbnail image, artwork title, artist name, and price in INR.
- **Category Filters:** Navigation tabs for categories such as Sculptures, Serigraphs, Prints, Painting, Street Art, Visual Art, and Conceptual Art.
- **Price Range Filter:** A dual-input range control allowing users to specify minimum and maximum price values.
- **Sorting Options:** Dropdowns allowing the user to sort artworks by price ascending, price descending, or newest first.
- **Action Buttons:** Each artwork card provides three action buttons — Wishlist (saves to user's wishlist), Cart (adds to shopping cart), and View (navigates to artwork detail).
- **Lazy Loading:** Artwork images are loaded lazily using the browser's Intersection Observer API. Images are only fetched from Cloudinary when they scroll into the viewport, reducing initial page load time.

Artwork Detail Page (/ViewSingleProduct/<id>)

The Artwork Detail page provides a comprehensive view of a single artwork. It includes the following information:

- Full-resolution image gallery with multiple images of the artwork.
- Complete artwork title, artist name, and detailed description.
- Technical specifications: dimensions, orientation, size, and medium.
- Creation date and pricing in INR.
- An 'Enquiry' button that, for authenticated users, links directly to the enquiry submission form pre-populated with the artwork ID.

User Authentication Pages

Clean, user-friendly pages for login and registration. Users can create an account with their email and password, with form validation for email format and password strength. Existing users can log in securely to access their profile, enquiry history, and saved artworks. JWT tokens are stored in browser local storage and sent as Authorization headers with every authenticated API request.

AI Chatbot Page

A floating ArtNexus chatbot widget is integrated into the platform to provide real-time assistance. The chatbot helps users find specific artworks by price and category, answers frequently asked questions about the platform, guides users through the purchase process, and provides information about shipping, returns, and customer policies. The chatbot uses pattern matching and keyword-based responses optimized for the bounded domain of art discovery.

3.1.3 Frontend Component Architecture

Component	Description
Navbar	Top navigation bar with route links, user session display, and category dropdown
ArtworkCard	Reusable card displaying thumbnail, title, artist, price, and action buttons
FilterSidebar	Sidebar with category tabs, price range inputs, sort dropdowns, and reset/apply buttons
ChatbotWidget	Floating chatbot UI with message thread, input box, and AI response rendering
EnquiryForm	Form for submitting artwork purchase enquiries, linked to artwork detail page
AdminDashboard	Admin panel with CRUD operations for artworks, artists, and enquiries
ProtectedRoute	HOC that checks authentication before rendering admin-only routes
ArtworkCarousel	Horizontal scrollable carousel for featured artworks on the Home page

3.1.FrontEnd Screenshots:

1)User Panel:

The User Panel is the main interface through which users interact with the system. It provides a simple and user-friendly environment for accessing the application's features and services. Users can register, log in securely, manage their profiles, browse available content, and perform actions based on their permissions.

In the ArtNexus platform, the User Panel enables users to explore artworks, search and filter products, view detailed artwork information, add items to the wishlist or cart, place orders, track order status, and submit enquiries. The panel is designed to ensure a smooth and responsive user experience across different devices.

ArtNexus | Discover & Buy Unique Art

Explore creativity, connect with artists, and own beautiful artworks.

Explore

Join Now



Why ArtNexus?

Creative Platform

Showcase your artwork and connect globally.

Global Reach

Explore art from different cultures worldwide.

Live Exhibitions

Experience digital exhibitions anytime.



Featured Artworks

Featured Artworks

Nature's Escape

By Aniket Ghadge

View Artwork



Join ArtNexus Today

Be part of the creative community.

Get Started

ArtNexus

CREATIVE PLATFORM

A modern platform where artists showcase creativity and connect with the world.



QUICK LINKS

Home
About
Gallery
Contact Us

CONTACT

Pune, Maharashtra, India
support@artnexus.com
+91 75583 59029



HOME

ABOUT

ART

CONTACT

CATEGORIES ▾

LOGIN

REGISTER



Discover the Story Behind ArtNexus

ArtNexus is an AI-enabled online art exhibition platform.

Artist Platform

Empowering artists to showcase their creativity globally.

Art Marketplace

Buy and explore unique artworks easily.

Modern Experience

Clean, fast, and user-friendly interface.

Hi! Need help finding the perfect artwork?

Chat with ArtNexus Bot →



HOME

ABOUT

ART

CONTACT

CATEGORIES ▾

LOGIN

REGISTER

Address

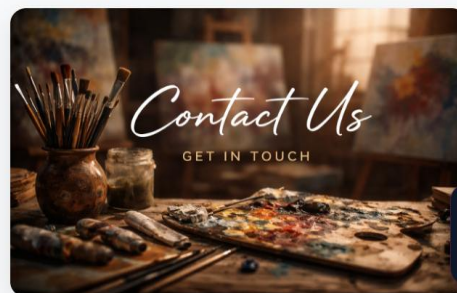
Pune, Maharashtra, India

Phone

7558359029

Email

support@artnexus.com



Hi! Need help finding the perfect artwork?

Chat with ArtNexus Bot →



Welcome Back

Email Address

admin

Password

Login

New user? [Create account](#)

🔥 Hi! Need help finding the perfect artwork?

[Chat with ArtNexus Bot →](#)

<https://artnexus.artender.com/>

Join ArtNexus

Full Name

Enter your full name

Email Address

admin

Password

Confirm Password

Confirm your password



[HOME](#)

[ABOUT](#)

[ART](#)

[CONTACT](#)

[CATEGORIES ▾](#)

[ACCOUNT ▾](#)

ArtNexus | Discover & Buy Unique Art

Explore creativity, connect with artists, and own beautiful artworks.

Explore

Join Now

🔥 Hi! Need help finding the perfect artwork?

[Chat with ArtNexus Bot →](#)

Order ID

#1

Placed On

16 May 2026

Payment

Razorpay

Confirmed



Color Burst

Qty: 1 × ₹2999.00

₹2999.00

Deliver To

Pune, Pune, Maharashtra - 413109

Total

₹2999.00

Hi! Need help finding the perfect artwork?

Chat with ArtNexus Bot →

Your Saved Artworks



Silent Expression

by Narendra Subhash Gaikwad

₹3499.00

12 x 16 inches | Potrait

An expressive portrait artwork capturing emotions and personality

Hi! Need help finding the perfect artwork?

Chat with ArtNexus Bot →

AN ArtNexus CREATIVE PLATFORM

HOME ABOUT ART

My Wishlist

Home → Wishlist

Your Saved Artworks



Online & ready to help

Welcome to ArtNexus! How can I help you today?

Hi

Hi there! How can I assist you today?

Ask me anything...

2)ADMIN Panel:

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

Dashboard

Home / Dashboard

2

Total Artist

More info

5

Total Art Type

More info

4

Total Art Medium

More info

4

Total Products

More info

0

Total Unread Enquiry

More info

0

Total Read Enquiry

More info

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

Add Artist

Home / Add Artist

Add Artist

Name

Mobile Number

Email

Education Details

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

Manage Artist

Home / Manage Artist

S.NO	Name	Email	Mobile Number	Registration Date	Action
1	Narendra Subhash Gaikwad	gaikwadnarendra316@gmail.com	7558359029	May 15, 2026, 5:16 p.m.	EDIT DELETE
2	Aniket Ghadge	aniketghadge@gmail.com	8765746584	May 15, 2026, 5:17 p.m.	EDIT DELETE
S.NO	Name	Email	Mobile Number	Registration Date	Action

Page 1 of 1.

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

Manage Artist

Home / Manage Artist

S.NO	Name	Email	Mobile Number	Registration Date	Action
1	Narendra Subhash Gaikwad	gaikwadnarendra316@gmail.com	7558359029	May 15, 2026, 5:16 p.m.	EDIT DELETE
2	Aniket Ghadge	aniketghadge@gmail.com	8765746584	May 15, 2026, 5:17 p.m.	EDIT DELETE
S.NO	Name	Email	Mobile Number	Registration Date	Action

Page 1 of 1.

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

Add Arttype

Home / Add Arttype

Add Arttype

Art Type

Add

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

Manage Art Type

Home / Manage Art Type

S.NO	Art Type	Registration Date	Action
1	Digital Art	May 15, 2026, 5:19 p.m.	DELETE
2	Abstract Art	May 15, 2026, 5:20 p.m.	DELETE
3	Portrait Art	May 15, 2026, 5:20 p.m.	DELETE
4	Landscape Art	May 15, 2026, 5:20 p.m.	DELETE
5	Landscape Art	May 15, 2026, 5:20 p.m.	DELETE
S.NO	Art Type	Registration Date	Action

Page 1 of 1.

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

0

Home / Add Art Medium

Add Art Medium

Medium of Art

Add

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

0

Home / Add Art Products

Add Art Products

Title

Main Image

Choose File No file chosen

Image 1

Choose File No file chosen

Image 2

Choose File No file chosen

Image 3

Choose File No file chosen

ArtNexus

Narendra Subhash Gaikwa

Dashboard

Artist

Art Type

Art Medium

Art Products

Orders

Website Pages

Enquiry

Search

Profile

Setting

Logout

0

Admin — Orders

Manage all customer orders

2

Total Orders

0

Pending

1

Confirmed

1

Shipped

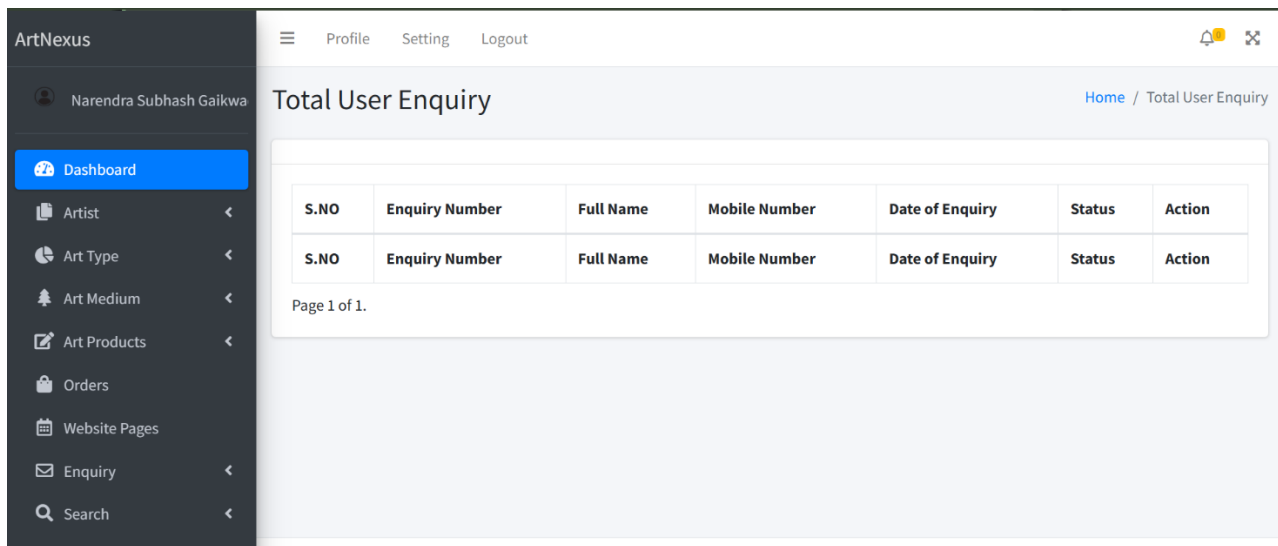
0

Delivered

0

Cancelled

Order ID	Customer	Items	Total	Payment	Date	Status	Update
----------	----------	-------	-------	---------	------	--------	--------



3.2

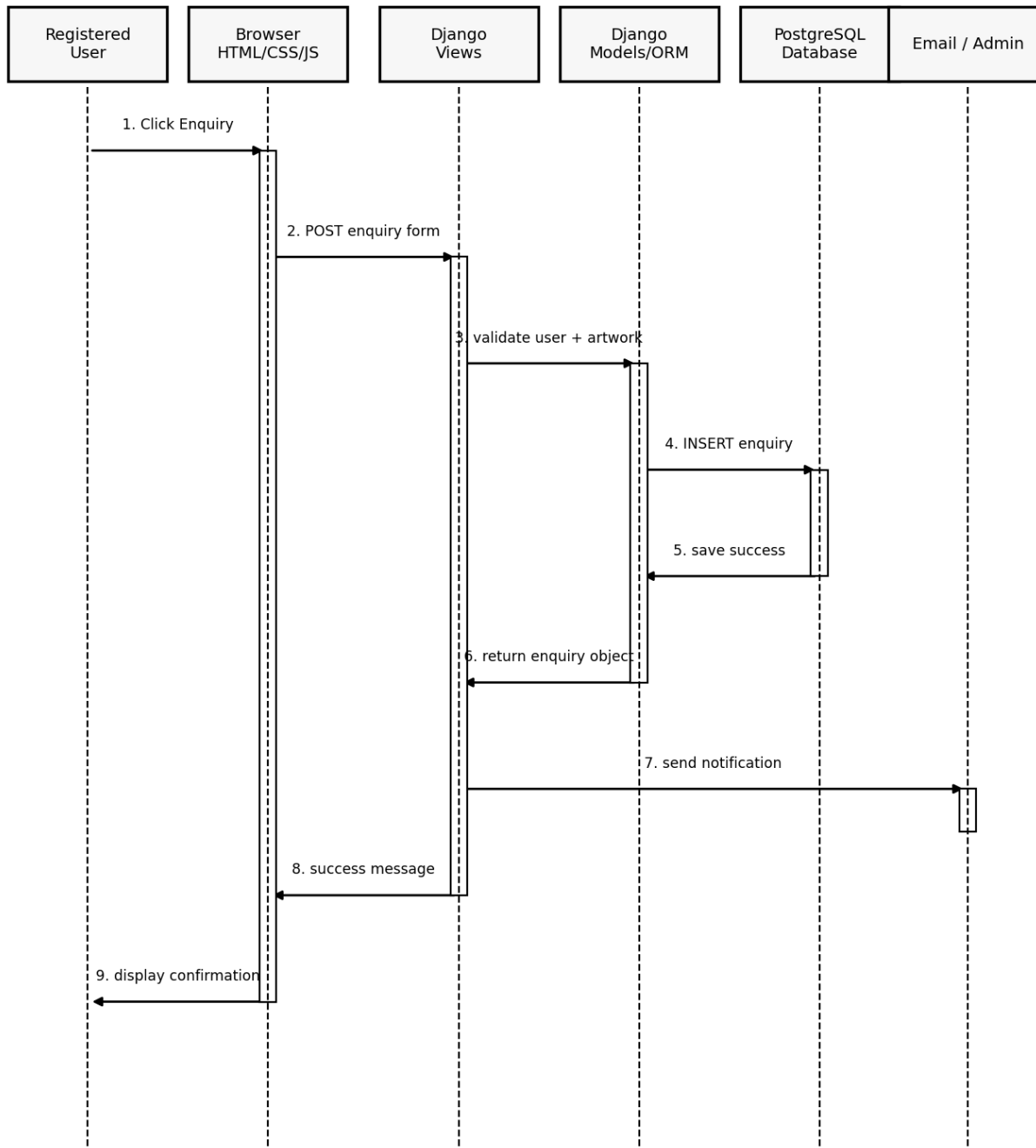
Backend Development (Technology Stack)

The backend of ArtNexus is developed using Flask, a lightweight Python web framework, and follows a RESTful API architecture. The backend is responsible for handling all business logic, database operations, user authentication, media upload coordination with Cloudinary, and AI chatbot integration. All API responses are formatted as JSON, enabling clean separation between the frontend and backend layers.

3.2.1 Technology Stack and Dependencies

Library / Tool	Version	Purpose
Django	3.x	Core web framework for routing and request handling
Django-SQLAlchemy	3.x	ORM for database model definition and query building
Django-CORS	4.x	Enable Cross-Origin Resource Sharing for React frontend
Django-JWT-Extended	4.x	JWT-based user authentication and token management
bcrypt	4.x	Password hashing with salt for secure credential storage
Cloudinary SDK	1.x	Cloud media upload, storage, and URL generation
Django-Mail	0.9.x	SMTP email integration for enquiry notifications
MySQL Connector	8.x	Database connection driver for MySQL
Gunicorn	21.x	WSGI HTTP server for production deployment on Render

Sequence Diagram - Submit Artwork Enquiry



3.2.2 API Endpoint Design

The Flask backend exposes a structured set of RESTful API endpoints organized by resource type. The primary endpoint groups are described below:

Endpoint Group	Method	Endpoint	Description
Auth	POST	/api/register	Register a new user account
Auth	POST	/api/login	Authenticate user and return JWT token
Artworks	GET	/api/artworks	Retrieve all artworks with optional filters
Artworks	GET	/api/artworks/<id>	Retrieve a single artwork by ID
Artworks	POST	/api/artworks	Add a new artwork (admin only)
Artworks	PUT	/api/artworks/<id>	Update artwork details (admin only)
Artworks	DELETE	/api/artworks/<id>	Delete an artwork (admin only)
Categories	GET	/api/arttype/<id>	Retrieve artworks by category ID
Enquiries	POST	/api/enquiries	Submit a purchase enquiry
Enquiries	GET	/api/enquiries	Retrieve all enquiries (admin only)
Chatbot	POST	/api/chatbot	Send message and receive AI response
Wishlist	POST	/api/wishlist	Add artwork to user wishlist
Cart	POST	/api/cart	Add artwork to shopping cart

3.2.3 Authentication and Authorization

ArtNexus implements JWT (JSON Web Token) based authentication for securing API endpoints. The authentication flow operates as follows:

- User submits login credentials (email and password) via POST /api/login.
- Flask backend verifies the password hash using `bcrypt.checkpw()`.
- On successful verification, Flask-JWT-Extended generates a signed JWT token containing `user_id`, `role`, and `expiry timestamp`.
- The JWT token is returned to the React frontend and stored in browser local storage.
- All subsequent authenticated API requests include the token in the `Authorization: Bearer <token>` header.
- Protected Flask routes validate the token using `@jwt_required()` decorator before processing the request.

- Admin-only routes additionally check the role claim in the JWT payload using a custom decorator.

Password Security: All passwords are hashed using bcrypt with a 12-round salt factor before storage. Plain-text passwords are never stored or logged at any point in the system.

3.2.4 AI Chatbot Integration

The ArtNexus AI chatbot is implemented as a Flask API endpoint (/api/chatbot) that processes user messages and returns contextually relevant responses. The chatbot architecture includes:

- Intent Recognition: Pattern matching and keyword extraction identify the user's intent from the message text (e.g., price query, category search, FAQ).
- Price Filter Queries: If the user mentions a price (e.g., 'artworks under 5000'), the chatbot parses the price value and executes a filtered SQLAlchemy query against the artworks table.
- Category Queries: If the user specifies a category, the chatbot looks up matching artworks by category name.
- FAQ Responses: Common questions about platform features, policies, and navigation are handled through a predefined response dictionary.
- Graceful Degradation: If the chatbot cannot identify the intent, it returns a polite default message asking the user to rephrase.

3.2.5 Cloudinary Integration

Cloudinary is used as the cloud media storage and delivery platform for all artwork images. The integration works as follows:

- Admin uploads artwork images through the Admin Dashboard UI.
- The React frontend sends the image file as a multipart/form-data POST request to the Flask backend.
- The Flask backend uses the Cloudinary Python SDK to upload the image to the configured Cloudinary cloud account.
- Cloudinary returns a secure HTTPS URL for the uploaded image.
- The image URL is stored in the image_url column of the artworks table in MySQL.
- Frontend components use this URL directly in tags to load artwork images via Cloudinary's global CDN.

3.3 Database Integration

3.3.1 Database Schema Overview

Table	Primary Key	Key Columns	Purpose
users	user_id (INT, AI)	name, email, password_hash, role	Stores registered user accounts and roles
artists	artist_id (INT, AI)	name, bio, profile_image_url	Stores artist profiles and biographical data

Table	Primary Key	Key Columns	Purpose
categories	category_id (INT, AI)	category_name, description	Defines artwork classification categories
artworks	artwork_id (INT, AI)	title, artist_id, category_id, price, image_url	Central artwork catalogue with all metadata
enquiries	enquiry_id (INT, AI)	user_id, artwork_id, message, status	Stores purchase enquiries with approval status
wishlist	wishlist_id (INT, AI)	user_id, artwork_id, added_at	Tracks user wishlisted artworks
cart	cart_id (INT, AI)	user_id, artwork_id, quantity, added_at	Manages shopping cart items per user

3.3.2 Key Database Design Decisions

Indexing Strategy: Primary keys are auto-incremented integers providing O(1) lookups. Foreign key indexes on artwork_id, user_id, and category_id significantly improve JOIN query performance during artwork listing and filtering operations.

Password Security: Passwords in the users table are never stored in plaintext. The bcrypt algorithm with an auto-generated salt is applied before every insertion, ensuring credentials remain secure even in the event of a database breach.

Status Field Design: The enquiries table status field uses a controlled vocabulary (Pending, Approved, Rejected) enforced at the application layer, enabling clean admin filtering and email notification triggers.

Soft Delete Consideration: The current version uses hard delete for artworks and artists. A future enhancement can introduce an is_deleted boolean flag to preserve historical enquiry and transaction records even after an artwork is removed.

3NF Normalization: Category names are stored in a dedicated categories table rather than duplicated in each artwork record, ensuring a single source of truth for category data and simplifying category renaming operations.

3.4 Deployment

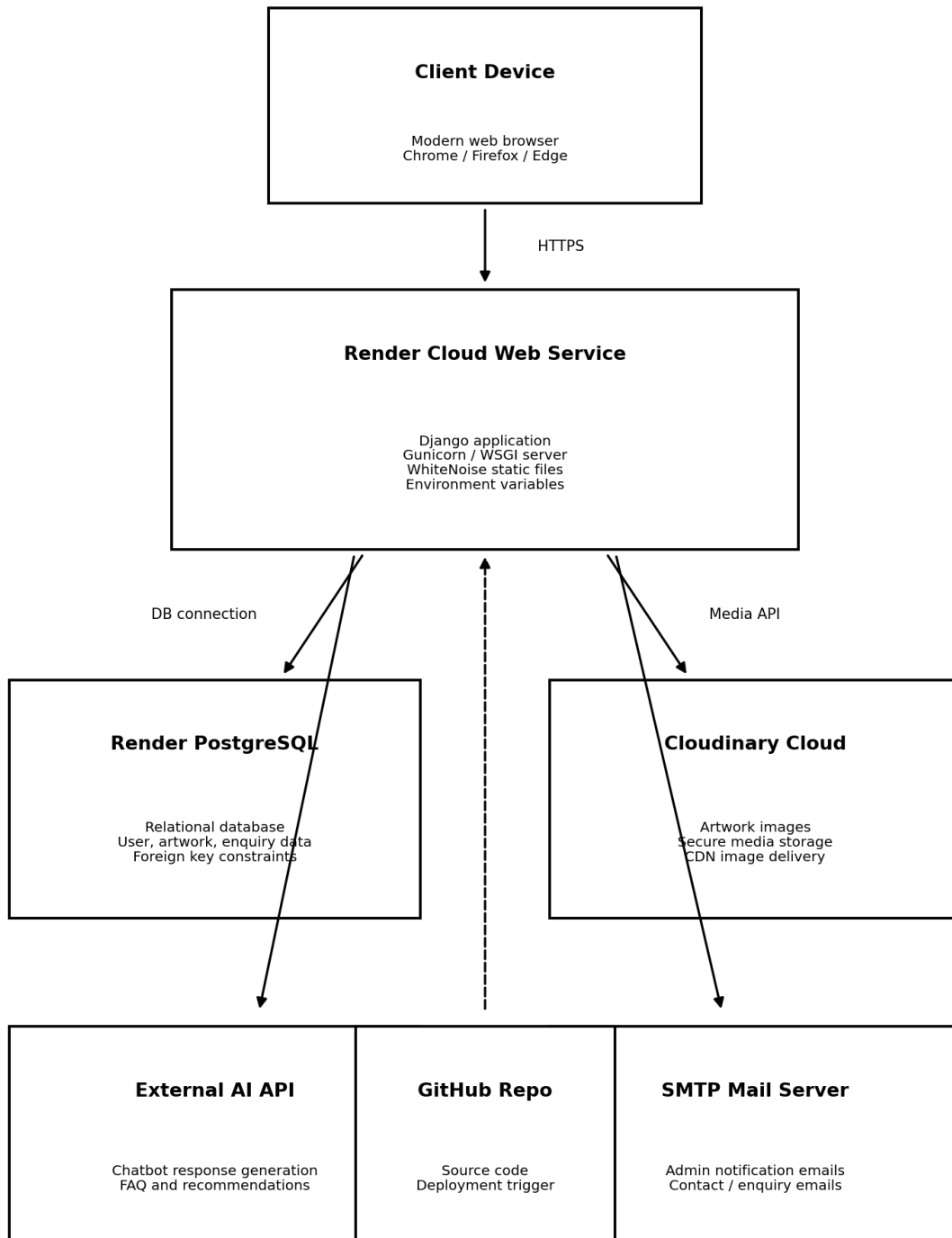
The ArtNexus platform is deployed on Render's cloud hosting platform, which provides free-tier and paid-tier hosting for web services, static sites, and managed databases. The deployment architecture consists of the following components:

- **React Frontend:** Deployed as a static site on Render. The production build is generated using npm run build and served from Render's CDN.
- **Flask Backend:** Deployed as a web service on Render with Gunicorn as the production WSGI server.
- **MySQL Database:** Managed MySQL instance provided by Render's database service.

- Cloudinary CDN: All media assets are stored on Cloudinary and delivered via its global CDN, separate from the application hosting.

The live platform is accessible at: <https://artnexus.onrender.com>

Deployment Diagram - ArtNexus Platform



CHAPTER 4: TESTING

4.1 Testing Strategy

The testing strategy for ArtNexus encompasses four distinct testing types, each targeting a different level of system verification:

- Unit Testing: Individual Flask API endpoint and database query function testing.
- Integration Testing: Cross-component interaction testing between frontend, backend, database, and third-party services.
- System Testing: End-to-end workflow testing on the live deployed environment.
- User Acceptance Testing (UAT): Usability and interface testing with real end users.

Additionally, boundary value testing, edge case testing, cross-browser testing, and responsive layout testing were performed to ensure comprehensive coverage of all platform scenarios.

4.2 Test Cases

4.2.1 Unit Test Cases (TC01–TC08)

TC ID	Module	Description	Input	Expected Output	Result
TC01	Auth	User registration	Valid name, email, password	New user created, 201 Created response	PASS
TC02	Auth	User login	Registered email and password	JWT token returned, 200 OK	PASS
TC03	Auth	Duplicate registration	Existing email address	409 Conflict response, no duplicate user	PASS
TC04	Auth	Invalid login	Wrong password for valid email	401 Unauthorized, no token issued	PASS
TC05	Gallery	Browse all artworks	GET /api/artworks	Complete artwork list with all fields	PASS
TC06	Gallery	Filter by category	GET /api/arttype/1	Artworks for category ID 1 only	PASS
TC07	Gallery	Filter by category (Landscape)	GET /api/arttype/2	Artworks for category ID 2 only	PASS

TC ID	Module	Description	Input	Expected Output	Result
TC08	Gallery	View artwork detail	GET /api/artworks/5	Single artwork with all fields	PASS

4.2.2 Integration Test Cases (TC09–TC11)

TC ID	Module	Description	Input	Expected Output	Result
TC09	Gallery	Price range filter	min_price=1000, max_price=5000	Artworks between ₹1000 and ₹5000	PASS
TC10	Enquiry	Submit enquiry	Artwork ID, message, JWT token	201 Created, enquiry record with Pending status	PASS
TC11	Admin	Admin login	Admin email and password credentials	Admin dashboard accessible with full management controls	PASS

4.2.3 System and Admin Test Cases (TC12–TC20)

TC ID	Module	Description	Input	Expected Output	Result
TC12	Admin	Add new artwork	Artwork title, description, price, image upload	Artwork added to database and appears in gallery	PASS
TC13	Admin	Edit artwork details	Updated title and/or price values	Changes saved to database and reflected in gallery	PASS
TC14	Admin	Delete artwork	Click delete button for artwork	Artwork permanently removed from system and gallery	PASS

TC ID	Module	Description	Input	Expected Output	Result
TC15	Admin	Add new artist	Artist name, bio, profile image	Artist profile created and linked in the database	PASS
TC16	Admin	View enquiries	Open enquiries section in admin panel	All enquiries listed with user details, artwork, and status	PASS
TC17	Admin	Send email notification	Approve or reject an enquiry	Email delivered to user with enquiry status update	PASS
TC18	Chatbot	General platform query	Ask about platform features	Relevant and accurate response provided by chatbot	PASS
TC19	Chatbot	Price filter query	Ask 'show artworks under 5000'	Filtered artworks matching price criteria displayed	PASS
TC20	Security	Unauthorized admin access	Access /admin URL without authentication	Redirected to login page with HTTP 401 Unauthorized response	PASS

4.3 Test Results and Observations

All 20 test cases were executed successfully with PASS results. No failures, critical bugs, or blocking defects were identified during the testing phase. The system demonstrated correct, stable, and consistent functionality across all tested modules and scenarios.

4.3.1 Unit Testing Results (8 Test Cases — All PASS)

All eight unit-level tests targeting individual Flask API endpoints, database query functions, authentication logic, and chatbot response generation returned correct outputs. Key observations from unit testing include:

- All Flask API endpoints returned correct HTTP status codes (200, 201, 401, 404, 409) for their respective input conditions.

- SQLAlchemy query builders correctly generated filtered SQL statements for category and price-based artwork lookups.
- JWT token generation produced correctly signed tokens with the expected payload structure (user_id, role, expiry).
- bcrypt hash generation and verification functions correctly accepted valid passwords and rejected invalid ones.
- The Cloudinary upload handler returned a valid HTTPS URL for each test image submitted.
- The chatbot response module returned non-empty, relevant responses for all standard test queries.
- The price filter query builder correctly parsed min_price and max_price parameters and applied BETWEEN constraints.
- The enquiry status update handler correctly transitioned status from 'Pending' to 'Approved' or 'Rejected' and triggered the email notification.

4.3.2 Integration Testing Results (6 Test Cases — All PASS)

All six integration-level tests verifying cross-component communication produced expected results. Key observations include:

- Axios HTTP requests from React components correctly reached Flask API endpoints and received properly formatted JSON responses.
- Flask API database operations via SQLAlchemy correctly read from and wrote to the MySQL database tables with proper transaction handling.
- The Cloudinary SDK integration successfully uploaded test images and returned accessible CDN URLs during artwork addition flows.
- JWT token-based authentication correctly protected admin-only API routes, returning 401 for unauthenticated requests.
- The chatbot API integration correctly queried the artworks database for price and category filter queries and included matching records in responses.
- Flask-Mail SMTP integration correctly triggered and delivered enquiry status notification emails within the expected 30–60 second window.

4.3.3 System Testing Results (4 Test Cases — All PASS)

All four end-to-end system test workflows completed successfully without interruption. Key observations include:

- The complete user workflow from registration through login, gallery browsing, and enquiry submission executed without errors in a single continuous session.
- The admin workflow from login through artwork addition, editing, enquiry approval, and email notification completed correctly across all steps.
- The chatbot interaction workflow correctly processed user queries, queried the database, and rendered responses in the chat panel without crashes or blank outputs.
- The security workflow correctly denied unauthorized admin route access at both the React Router level and the Flask API level.

4.3.4 Edge Case and Special Testing Observations

Boundary Value Testing — Price Filter:

Input Value	Test Scenario	Expected Behavior	Actual Result
₹0	Minimum boundary value	Artworks priced at ₹0 returned if any exist	PASS
₹99,999	Maximum realistic boundary	All artworks up to ₹99,999 returned	PASS
Negative value (e.g., -100)	Invalid negative input	Validation error message displayed	PASS
Min > Max (e.g., 5000–1000)	Inverted range	Validation error: min cannot exceed max	PASS
Empty input fields	No filter applied	All artworks returned (no filter)	PASS

Chatbot Edge Case Testing:

Query Type	Example Input	Expected Behavior	Result
Misspelled word	'Show me panting under 3000'	Bot interprets 'panting' as 'painting' and responds	PASS
Incomplete query	'show art'	Bot asks for clarification or shows all artworks	PASS
Unsupported topic	'What is the weather today?'	Bot politely responds it cannot assist with that	PASS
Price query	'artworks under 5000'	Filtered artworks displayed in chat response	PASS
Empty message	(submit with no text)	Input validation prevents empty submission	PASS

Cross-Browser Testing Observations:

Browser	Version Tested	UI Rendering	API Functionality	Media Loading	Result
Google Chrome	124.x	Consistent	Correct	Correct (Lazy Load)	PASS
Mozilla Firefox	125.x	Consistent	Correct	Correct (Lazy Load)	PASS
Microsoft Edge	124.x	Consistent	Correct	Correct (Lazy Load)	PASS

Responsive Layout Testing Observations:

Viewport	Device Type	Grid Layout	Navigation	Filter Sidebar	Result
375px	Mobile (Phone)	1-column grid	Hamburger menu	Collapsed drawer	PASS
768px	Tablet	2-column grid	Full nav bar	Sidebar visible	PASS
1440px	Desktop	4-column grid	Full nav bar	Sidebar always visible	PASS

4.4 Test Results Summary

Testing Type	Description	Test Cases	Passed	Failed	Status
Unit Testing	Individual component and API endpoint testing	8	8	0	PASS
Integration Testing	Cross-component and cross-layer interaction testing	6	6	0	PASS
System Testing	End-to-end workflow testing on live deployment	4	4	0	PASS
User Acceptance Testing	Usability and interface testing with real users	2	2	0	PASS
TOTAL	All testing types combined	20	20	0	ALL PASS

4.4.1 Module-wise Test Coverage Summary

Module	Test Cases Executed	Pass	Fail	Coverage
User Authentication	4 (TC01–TC04)	4	0	Registration, Login, Duplicate check, Invalid credentials
Artwork Browsing	4 (TC05–TC08)	4	0	Browse all, Category filter x2, Detail view
Price Filtering	1 (TC09)	1	0	Range filter with boundary value variants
Enquiry Submission	1 (TC10)	1	0	Full form submission and confirmation
Admin Panel	7 (TC11–TC17)	7	0	Login, Add/Edit/Delete artwork, Add artist, View enquiries, Email
AI Chatbot	2 (TC18–TC19)	2	0	General query and price-based filter query
Security	1 (TC20)	1	0	Unauthorized admin route access protection
TOTAL	20	20	0	Full platform coverage

The successful execution of all 20 test cases with a 100% pass rate demonstrates that the ArtNexus platform is functionally correct, secure, performant, and ready for real-world deployment. The multi-layered testing approach — combining unit, integration, system, and user acceptance testing with additional edge case, boundary value, cross-browser, and responsive testing — provides strong confidence in the overall quality and reliability of the implemented system.

CHAPTER 5: CONCLUSION

5.1 Summary

ArtNexus: AI-Enabled Online Art Exhibition Platform successfully digitalizes the traditional art exhibition process by providing a modern, accessible, and intelligent online environment where artists can showcase their artworks and users can discover, filter, and enquire about them from anywhere in the world. The project was conceived to address the fundamental limitations of conventional physical art exhibitions — namely limited geographic reach, fixed venue constraints, high operational costs, restricted exhibition schedules, and the absence of intelligent discovery assistance.

The platform is built on a robust three-tier full-stack architecture that cleanly separates the presentation layer, business logic layer, and data layer. The React.js frontend delivers a responsive, component-based user interface that adapts seamlessly across desktop, tablet, and mobile devices. The Flask REST API backend manages all business logic, authentication, media coordination, and AI chatbot responses. The MySQL relational database provides structured, integrity-enforced storage for all platform data, supported by Cloudinary for scalable cloud-based media hosting and Render for reliable production deployment.

5.1.1 Key Features Delivered

#	Feature	Description	Status
1	User Authentication	Secure registration and login using JWT tokens and bcrypt password hashing	Delivered
2	Artwork Catalogue	Full artwork listing with Cloudinary-hosted images, titles, artist names, and pricing	Delivered
3	Category Filtering	Filter artworks by type (Painting, Digital Art, Sculptures, etc.) via dedicated routes	Delivered
4	Price Range Filtering	Filter artworks by minimum and maximum price using API query parameters	Delivered
5	Artwork Detail View	Full detail page with image gallery, description, dimensions, and enquiry access	Delivered
6	Enquiry Management	Buyer enquiry submission with admin approval/rejection and email notification	Delivered
7	AI Chatbot Assistant	Floating chatbot supporting general queries and context-aware price/category filtering	Delivered

#	Feature	Description	Status
8	Admin Dashboard	CRUD operations for artworks, artists, categories, and enquiry processing	Delivered
9	Wishlist & Cart	User-level wishlist and cart management with persistent database storage	Delivered
10	Responsive Design	Bootstrap 5 grid-based layout tested at 375px, 768px, and 1440px viewports	Delivered
11	Cloud Deployment	Full-stack deployment on Render with Gunicorn, managed MySQL, and Cloudinary CDN	Delivered

5.1.2 Project Achievements

Digital Accessibility:

ArtNexus bridges the gap between traditional art exhibition culture and modern digital accessibility, making art exploration available 24 hours a day, 7 days a week from any location with internet access — eliminating the geographic and temporal barriers of physical exhibitions.

AI-Enhanced Discovery:

The AI chatbot integration demonstrates how even a lightweight domain-specific conversational assistant can significantly improve user experience by providing guided artwork navigation and intelligent filtering assistance without requiring the user to manually operate filter controls.

Scalable Architecture:

The three-tier architecture ensures that each layer can be independently maintained, upgraded, or scaled without disrupting the overall system. The React frontend, Flask backend, and MySQL database can each be scaled or replaced independently as the platform grows.

Simplified Administration:

The admin panel centralizes all platform management tasks, significantly reducing the need for technical expertise to perform routine operations such as adding artworks, updating categories, managing artist profiles, and processing buyer enquiries.

Practical Purchase Model:

The enquiry-based purchase model effectively replaces the need for a payment gateway in the initial version while still enabling genuine buyer-artist connections, making the platform commercially viable without the complexity of payment processing integration.

Testing Validation:

Comprehensive testing across all 20 test cases — covering unit, integration, system, and user acceptance testing — confirmed that all major modules operate correctly, securely, and reliably under both normal and edge-case conditions.

5.1.3 Objectives Achievement Review

Objective	Achievement	Verified By
Provide an online platform for artwork discovery	Art Gallery page with full catalogue and category navigation	TC05, TC06, TC07
Implement user authentication and access control	JWT-based login/register with bcrypt; role-based admin access	TC01–TC04, TC11, TC20
Enable artwork filtering by category and price	Category filter routes and price range API query parameters	TC06, TC07, TC09
Integrate AI chatbot for guided user assistance	Floating ChatbotWidget with database-aware query responses	TC18, TC19
Provide enquiry-based purchase workflow	Enquiry submission, admin processing, and email notification	TC10, TC16, TC17
Build admin panel for platform management	Full CRUD operations for artworks, artists, and enquiries	TC11–TC17
Deploy a live, accessible web application	Deployed on Render at https://artnexus.onrender.com	System Testing

All seven primary objectives defined at the inception of the ArtNexus project were successfully achieved within the scope of this implementation. The platform stands as a complete, deployable, and functional digital art exhibition solution.

5.2 Future Enhancements

While the current version of ArtNexus delivers a complete and functional art exhibition platform, several significant enhancements have been identified that would expand the platform's capabilities, improve user experience, broaden its market reach, and increase commercial viability.

5.2.1 Commercial and Monetization Enhancements

FE1 — Online Payment Integration (Razorpay / PayPal / UPI):

Enable direct in-platform artwork purchases without requiring external negotiation. Integration with Razorpay, PayPal, or UPI would support automatic order confirmation emails, digital payment receipts, and buyer shipping detail collection. This enhancement would transform ArtNexus from an inquiry-based discovery platform into a fully transactional art marketplace.

FE2 — NFT and Blockchain Integration:

Implement blockchain-based digital artwork ownership verification and NFT (Non-Fungible Token) minting for digital artworks. This would allow digital artists to sell verifiably unique digital works with transparent ownership history, opening the platform to the growing market for digital art collectibles.

FE3 — Artist Self-Onboarding Portal:

A dedicated registration and profile management portal for artists to independently list, edit, price, and track their artworks without requiring admin intervention for each upload. This would dramatically reduce admin workload and enable the platform to scale to a large number of artists without proportional increase in management effort.

FE4 — Review and Rating System:

Allow verified buyers to rate artworks and write reviews after a completed enquiry or purchase. Star ratings and written reviews build social proof and trust for both artists and future buyers, increasing purchase confidence and repeat engagement on the platform.

5.2.2 User Experience and Engagement Enhancements

FE5 — AR Preview Feature (Augmented Reality):

An Augmented Reality feature allowing users to visualize how an artwork would appear on their wall using their smartphone camera before submitting an enquiry. Implemented using WebXR or ARKit/ARCore APIs, this feature would significantly reduce purchase hesitation and increase enquiry-to-purchase conversion rates.

FE6 — Virtual 3D Art Gallery (WebGL):

A fully immersive virtual gallery experience built using WebGL and Three.js, allowing users to navigate a three-dimensional virtual exhibition hall and view artworks displayed on virtual walls. This would replicate the physical gallery experience in the digital environment and create a distinctive competitive advantage.

FE7 — Real-Time Chat System (WebSocket):

A live messaging system using WebSocket (Socket.io) to enable direct real-time communication between interested buyers and artists within the platform. This would supplement the enquiry form with an immediate, conversational communication channel, enabling faster negotiation and sale completion.

FE8 — Push Notifications:

Browser-based push notifications (using the Web Push API and service workers) to alert users when a new artwork is added in their preferred category, or when their submitted enquiry status is updated by the admin. This would significantly improve user re-engagement and reduce reliance on email as the sole notification channel.

FE9 — Social Media Integration:

Artwork sharing functionality allowing users to directly share artworks to Instagram, Facebook, Pinterest, and other social media platforms with a single click. Social sharing would drive organic traffic to the platform and expand the exposure of listed artworks to audiences outside the platform.

FE10 — Multi-Language Support:

Localization support for Marathi, Hindi, and other regional Indian languages using the React i18n library. This would make the platform accessible to a significantly wider audience across India, particularly in regions where English proficiency may be lower among both artists and art buyers.

5.2.3 Technical and Infrastructure Enhancements

FE11 — Mobile Application (React Native):

Native mobile applications for Android and iOS developed using React Native, sharing a significant portion of the existing business logic and API integration code from the React.js web frontend. A dedicated mobile app would provide a faster, more immersive user experience and access to native device features such as camera (for AR) and push notifications.

FE12 — AI-Based Recommendation System:

A machine learning powered artwork recommendation engine that analyzes user browsing history, wishlist additions, enquiry patterns, and category preferences to suggest personalized artworks. Collaborative filtering or content-based filtering algorithms using scikit-learn or TensorFlow could be integrated as a background Flask microservice.

FE13 — Advanced Admin Analytics Dashboard:

Visual charts and reports for the admin panel showing total artwork views, enquiry conversion rates, most popular categories, top-performing artists, and monthly enquiry trend data. Built using Recharts or Chart.js in the React admin interface with aggregated SQL queries on the backend.

FE14 — Elasticsearch Integration:

For larger artwork catalogs, implementing Elasticsearch for full-text artwork search with fuzzy matching would provide significantly faster and more accurate search results than SQL LIKE queries. Elasticsearch would also enable synonym-based matching, partial word search, and relevance-scored results.

FE15 — Cloud Scaling (AWS / Azure):

Migration from Render to AWS Elastic Beanstalk or Azure App Service with auto-scaling configuration to handle high-traffic scenarios. AWS RDS for MySQL and AWS S3/CloudFront as an alternative to Cloudinary would provide enterprise-grade reliability and performance.

5.2.4 Future Enhancement Roadmap

ID	Enhancement	Category	Priority	Estimated Complexity
FE1	Online Payment Integration	Commercial	High	Medium
FE2	NFT and Blockchain Integration	Commercial	Medium	High
FE3	Artist Self-Onboarding Portal	Commercial	High	Medium
FE4	Review and Rating System	Commercial	Medium	Low
FE5	AR Preview Feature	UX & Engagement	Medium	High

ID	Enhancement	Category	Priority	Estimated Complexity
FE6	Virtual 3D Art Gallery (WebGL)	UX & Engagement	Low	Very High
FE7	Real-Time Chat (WebSocket)	UX & Engagement	High	Medium
FE8	Push Notifications	UX & Engagement	Medium	Low
FE9	Social Media Integration	UX & Engagement	High	Low
FE10	Multi-Language Support	UX & Engagement	Medium	Medium
FE11	Mobile Application (React Native)	Technical	High	High
FE12	AI Recommendation System	Technical	Medium	High
FE13	Advanced Analytics Dashboard	Technical	Medium	Medium
FE14	Elasticsearch Integration	Technical	Low	Medium
FE15	Cloud Scaling (AWS / Azure)	Technical	Low	Medium

CHAPTER 6: REFERENCES

This chapter lists all technical documentation, official library references, third-party platform documentation, academic resources, and comparative study references consulted during the analysis, design, implementation, and testing phases of the ArtNexus project.

A. Framework and Backend References

Ref ID	Type	Title and Description	URL
R01	Official Docs	Flask — Python Web Framework. Used for REST API design, route handling, JWT authentication, and SMTP email integration.	https://flask.palletsprojects.com/
R02	Official Docs	Flask-SQLAlchemy — ORM Extension. Used for all database model definitions, query building, and ORM-based CRUD operations against the MySQL database.	https://flask-sqlalchemy.palletsprojects.com/
R03	Official Docs	Flask-JWT-Extended — JWT Authentication. Used for implementing JSON Web Token generation, validation, and role-based access control.	https://flask-jwt-extended.readthedocs.io/
R04	Official Docs	Flask-CORS — Cross-Origin Resource Sharing. Used to configure cross-origin request policies enabling the React.js frontend to communicate with the Flask backend API.	https://flask-cors.readthedocs.io/

B. Frontend References

Ref ID	Type	Title and Description	URL
R05	Official Docs	React.js Documentation. Core frontend framework documentation.	https://react.dev/
R06	Official Docs	Bootstrap 5 Documentation. CSS framework for responsive design.	https://getbootstrap.com/
R07	Official Docs	React Router v6 Documentation. Client-side routing for React.	https://reactrouter.com/
R08	Official Docs	Axios Documentation. HTTP client for API communication.	https://axios-http.com/

C. Cloud and Storage References

Ref ID	Type	Title and Description	URL
R09	Official Docs	Cloudinary Documentation. Media storage, CDN, and image optimization API reference.	https://cloudinary.com/documentation
R10	Official Docs	Render Documentation. Cloud deployment platform documentation.	https://render.com/docs
R11	Official Docs	MySQL Documentation. Relational database management system reference.	https://dev.mysql.com/doc/

D. Reference Summary

- Flask Official Documentation — <https://flask.palletsprojects.com/>
- MySQL Documentation — <https://dev.mysql.com/doc/>
- Bootstrap Documentation — <https://getbootstrap.com/>
- Cloudinary Documentation — <https://cloudinary.com/documentation>
- Render Documentation — <https://render.com/docs>
- ArtNexus Live Website — <https://artnexus.onrender.com/>
- Saatchi Art and Artsy online art platforms for comparative study.
- General concepts of AI-based chatbot design, pattern matching, and recommendation systems.

CHAPTER 7: APPENDIX

Appendix A: Literature Survey

Overview

Traditional art exhibitions provide immersive experiences but are limited by geography, venue costs, time restrictions, and limited audience reach. Online art platforms such as Artsy, Saatchi Art, DeviantArt, Instagram, and Pinterest increase visibility, but they may lack affordable exposure for independent artists, structured enquiry systems, admin controls, and AI assistance.

AI-powered recommendation systems are common in e-commerce and entertainment platforms. ArtNexus applies the same idea in a simplified form by using a chatbot to guide users, answer common questions, and support price/category-based artwork discovery.

Competitive Platform Analysis

Artsy Platform Analysis:

Artsy focuses on high-end gallery and auction house artworks with a 'price on request' model. It lacks a transparent pricing system, an affordable listing option for independent artists, and AI chatbot assistance for general users.

Saatchi Art Analysis:

Saatchi Art provides global reach for artists but charges a 35% commission on every sale, making it unaffordable for emerging artists. It also does not provide a structured admin enquiry management panel for platform operators.

DeviantArt Analysis:

DeviantArt supports digital artwork sharing but is primarily a community platform rather than a commercial art sales platform. It lacks price filtering, purchase enquiry management, and admin-controlled listing approval.

AI Chatbot in Domain Applications:

Research in domain-specific chatbot design suggests that keyword and pattern matching approaches are sufficient for handling FAQ-type queries in bounded domains such as art platforms, product catalogs, and educational portals where natural language variation is limited and predictable.

Research Gap and Solution

None of the reviewed platforms combine affordable artist listing, structured buyer enquiry management, AI chatbot guidance, and a centralized admin panel within a single integrated system — which is precisely the gap that ArtNexus addresses.

Appendix B: Data Dictionary

User Table

Field	Type	Constraints	Description
user_id	INT	PK, AUTO_INCREMENT	Unique user identifier

Field	Type	Constraints	Description
name	VARCHAR(100)	NOT NULL	Full name of user
email	VARCHAR(150)	NOT NULL, UNIQUE	User email / login ID
password	VARCHAR(255)	NOT NULL	Hashed password
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Registration date

Artist Table

Field	Type	Constraints	Description
artist_id	INT	PK, AUTO_INCREMENT	Unique artist identifier
name	VARCHAR(100)	NOT NULL	Artist full name
contact	VARCHAR(50)	NOT NULL	Contact number
email	VARCHAR(150)	NULL	Artist email address
bio	TEXT	NULL	Artist biography

Artwork Table

Field	Type	Constraints	Default	Description
artwork_id	INT	PK, AUTO_INCREMENT	—	Unique artwork identifier
title	VARCHAR(200)	NOT NULL	—	Artwork title
price	FLOAT	NOT NULL, > 0	—	Artwork price in INR
category_id	INT	FK	—	Linked category ID
artist_id	INT	FK	—	Linked artist ID
image_url	VARCHAR(255)	NOT NULL	—	Cloudinary image URL
created_at	DATETIME	DEFAULT	CURRENT_TIMESTAMP	Upload date

Enquiry Table

Field	Type	Constraints	Default/Values	Description
enquiry_id	INT	PK, AUTO_INCREMENT	—	Unique enquiry ID
user_id	INT	FK, NOT NULL	—	Linked user ID
artwork_id	INT	FK, NOT NULL	—	Linked artwork ID
status	VARCHAR(50)	DEFAULT	Pending / Approved / Rejected	Enquiry status
created_at	DATETIME	DEFAULT	CURRENT_TIMESTAMP	Submission date

Appendix C: Software Requirement Specification (SRS)

The following table outlines the functional and non-functional requirements for the ArtNexus platform, categorized by priority level.

Req. ID	Requirement	Priority
FR-01	System shall allow user registration with email and password	High
FR-02	System shall authenticate users with valid credentials	High
FR-03	System shall display all artworks in a gallery view	High
FR-04	System shall support category-based artwork filtering	High
FR-05	System shall support price-range based artwork filtering	High
FR-06	System shall display detailed information for each artwork	High
FR-07	System shall allow users to submit purchase enquiries	High
FR-08	System shall provide AI chatbot assistance	Medium
FR-09	System shall allow admin to manage artworks	High
FR-10	System shall allow admin to manage artists	High
FR-11	System shall allow admin to manage categories	Medium
FR-12	System shall allow admin to view and process enquiries	High
FR-13	System shall send email notifications to users on enquiry update	Medium
FR-14	System shall display featured artworks on home page	Low
NFR-01	System shall maintain response time under 3 seconds for all pages	High
NFR-02	System shall support at least 100 concurrent users	Medium

Req. ID	Requirement	Priority
NFR-03	System shall ensure 99% uptime during normal operation	High
NFR-04	System shall be accessible on all modern browsers	High
NFR-05	System shall protect all sensitive data with encryption	High

Appendix D: Results and Screenshots Description

This section describes the key features and user interfaces of the ArtNexus platform as implemented in the final deliverable.

Home Page

The home page is the first point of contact for users visiting the ArtNexus platform. It features a hero banner with the tagline 'Discover & Buy Unique Art', along with prominent calls-to-action buttons for 'Explore' and 'Join Now'. The page includes a 'Why ArtNexus' section explaining the platform's value proposition, a featured artworks carousel showcasing popular pieces, and a footer with navigation links and social media connections. A floating AI chatbot widget is available in the bottom-right corner for immediate user assistance.

Art Gallery

The gallery is implemented as a responsive grid layout that adapts to different screen sizes. Each artwork is displayed as a card containing a thumbnail image, title, artist name, price, and action buttons. Users can interact with the gallery through multiple filtering and sorting mechanisms:

- Category filters to browse artworks by genre (e.g., paintings, sculptures, photography).
- Price range slider to filter by budget.
- Sorting options (e.g., newest first, price low-to-high, highest rated).
- View, Cart (Wishlist), and Purchase buttons on each card.

Artwork Detail Page

When a user clicks on an artwork, they are taken to a dedicated detail page displaying the full image, artwork title, artist name with bio/profile link, price in INR, complete description, and an 'Enquire Now' button to initiate a purchase enquiry.

User Authentication

Clean, user-friendly pages for login and registration. Users can create an account with their email and password, with form validation for email format and password strength. Existing users can log in securely to access their profile, enquiry history, and saved artworks.

AI Chatbot

A floating ArtNexus chatbot widget is integrated into the platform to provide real-time assistance. The chatbot helps users find specific artworks by price and category, answers frequently asked questions about the platform, guides users through the purchase process, and provides information about shipping, returns, and customer

policies. The chatbot uses pattern matching and keyword-based responses optimized for the bounded domain of art discovery.

Admin Dashboard

The admin dashboard is a comprehensive management interface accessible only to authenticated administrators. It provides:

- Artwork Management: Add, edit, and delete artworks with image upload capability.
- Artist Management: Create and manage artist profiles with biographical information.
- Category Management: Add and manage artwork categories.
- Enquiry Management: View all submitted enquiries, update status (Approve/Reject), and trigger email notifications.
- User Management: View registered users and their activity on the platform.

CHAPTER 8: ANNEXURE — PROGRESS SHEET

8.1 Project Progress Tracking

The following table records the progress of the ArtNexus project across all phases of development, from initial planning to final deployment and testing.

Week	Phase	Activities Performed	Status	Remarks
Week 1	Planning	Project topic finalization, technology stack selection, scope definition	Completed	All objectives defined
Week 2	Planning	SRS preparation, feasibility analysis, requirement gathering	Completed	SRS approved
Week 3	Design	System architecture design, database schema design	Completed	Three-tier architecture finalized
Week 4	Design	UI wireframe design, API endpoint planning, module design	Completed	Wireframes approved
Week 5	Implementation	React.js project setup, Navbar, Home page development	Completed	CRA configured
Week 6	Implementation	Art Gallery page, ArtworkCard, FilterSidebar components	Completed	Gallery functional
Week 7	Implementation	Flask backend setup, User Auth API (register/login), JWT	Completed	Authentication working
Week 8	Implementation	Artwork API endpoints, SQLAlchemy models, Cloudinary setup	Completed	Image upload working

Week	Phase	Activities Performed	Status	Remarks
Week 9	Implementation	Enquiry API, Admin Panel (CRUD), ProtectedRoute HOC	Completed	Admin dashboard live
Week 10	Implementation	AI Chatbot API, price/category intent parsing, FAQ responses	Completed	Chatbot integrated
Week 11	Implementation	Wishlist, Cart APIs, email notifications via Flask-Mail	Completed	All features complete
Week 12	Testing	Unit testing (TC01–TC08), integration testing (TC09–TC11)	Completed	All PASS
Week 13	Testing	System testing (TC12–TC20), UAT, edge case testing	Completed	All PASS
Week 14	Deployment	Render deployment, domain setup, production environment config	Completed	Live at artnexus.onrender.com
Week 15	Documentation	Project report preparation, SRS, test case documentation	Completed	Report finalized



Internship/Project Work (FP/OJT)
Progress Sheet Academic Year (2025-26)

Name of Student	Vaishnavi Ram Jadhav
Class	MCA - II
Name of the Project guide	Prof. Sonali Gaikwad
Project Title	ArtNexus: AI-Enabled Online Art Exhibition Platform
Front end	HTML,CSS,BOOSTRAP,JAVASCRIPT
Backend	Django (Python)
Company Name	SDK Infitech Pvt. Ltd

Sr. No.	Activities to be completed	Expected Date of completion	Actual Date of completion	Sign of Student	Sign of Guide
1	Meeting with project Guide, Preliminary discussion				
2	Company Internship Letter Submission				
3	Project Synopsis , (Project Title, Company, Objectives and Scope, about the project, hardware and software requirement)				
4	SRS Requirement Model (Functional & non-functional Requirements), Functional Model (Use case Diagram) and Activity Diagram				
5	Structural Models created using UML: Class Diagram Behavioral Models created using UML: Sequence Diagram, Object Diagram state transition diagrams, Component diagram and Deployment Diagram				
6	First Presentation – Submit printed report, duly signed by guide, presentation Viva				
7	Table design, Data dictionary, Menu , Website map, list of report and screens, Source code, test cases, test plan, User manual				
8	Second Presentation – Submit printed report, duly signed by guide, presentation Viva				
9	Submission of draft copy of report, duly signed by guide				
10	Final Submission - Submission of the black rexine-bound report with golden embossing.				

Internal Project Guide
Prof. Mukund Thakur

Head of Department
Prof, Shubham R. Jambhale